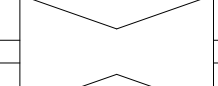
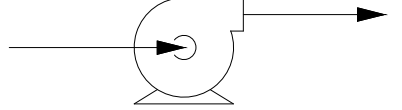
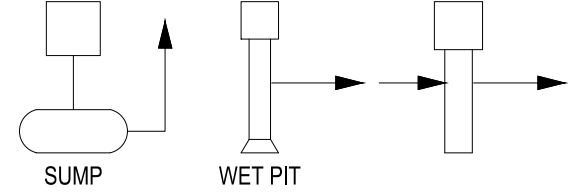
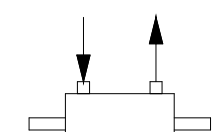
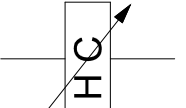
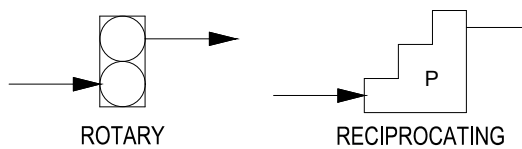
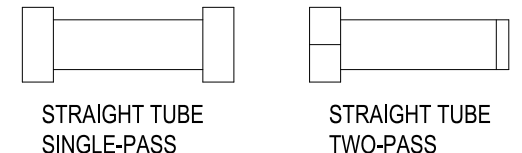
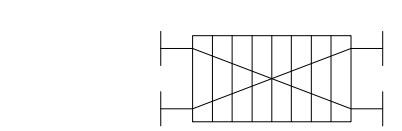
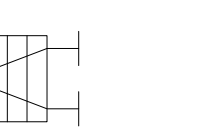
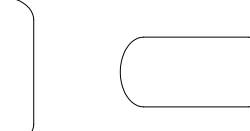
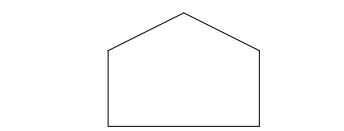
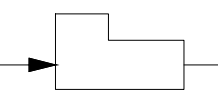
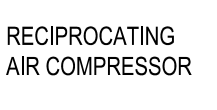
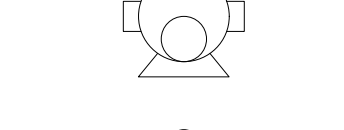
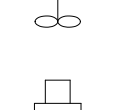
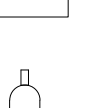
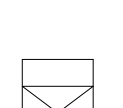
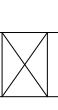
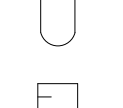
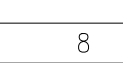
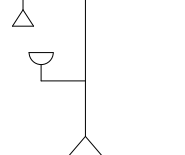
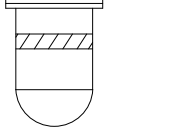
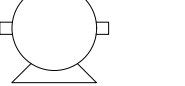
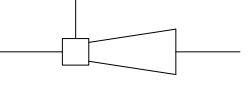
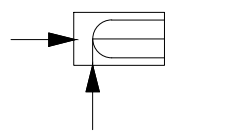
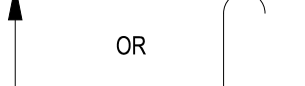
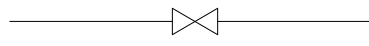
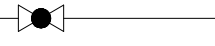
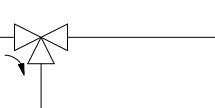
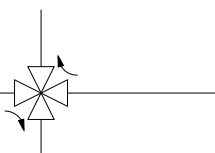
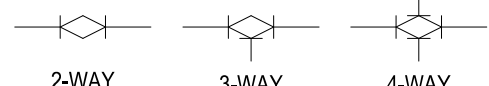
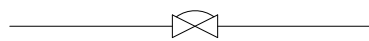
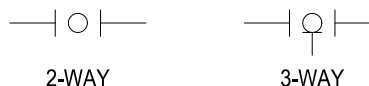
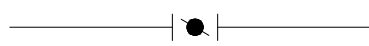
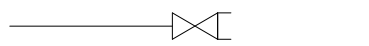
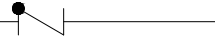
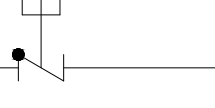
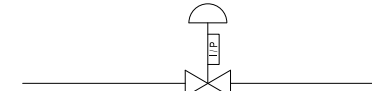
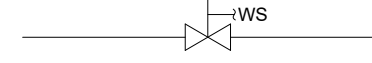
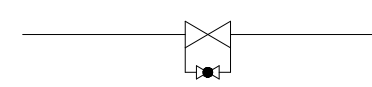
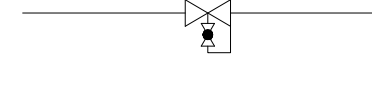
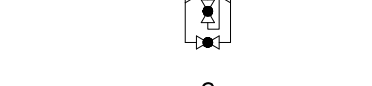
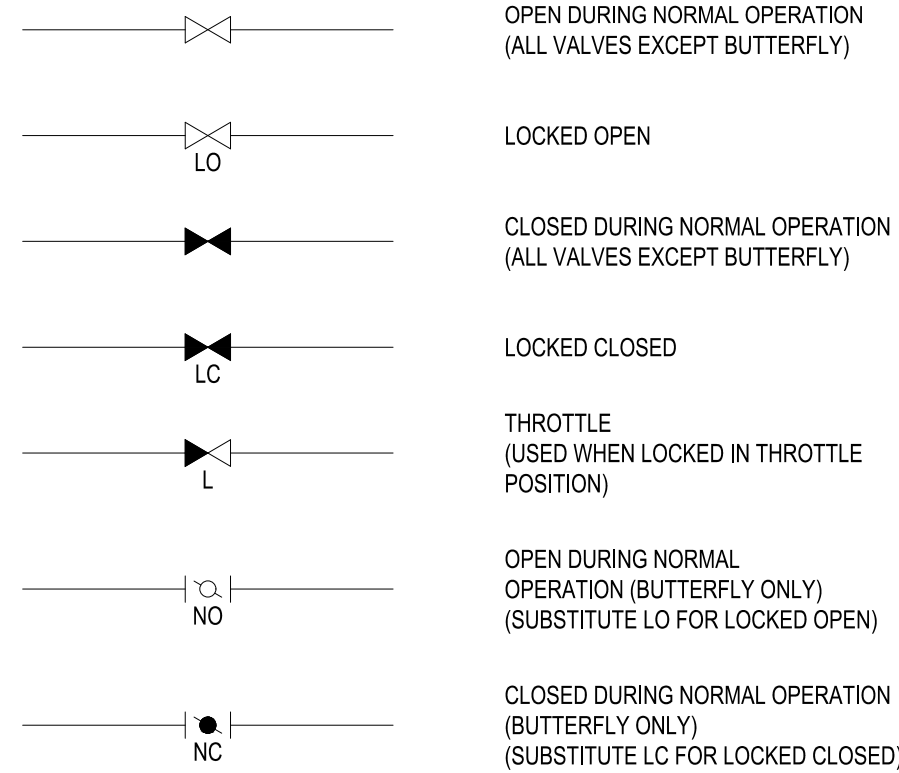
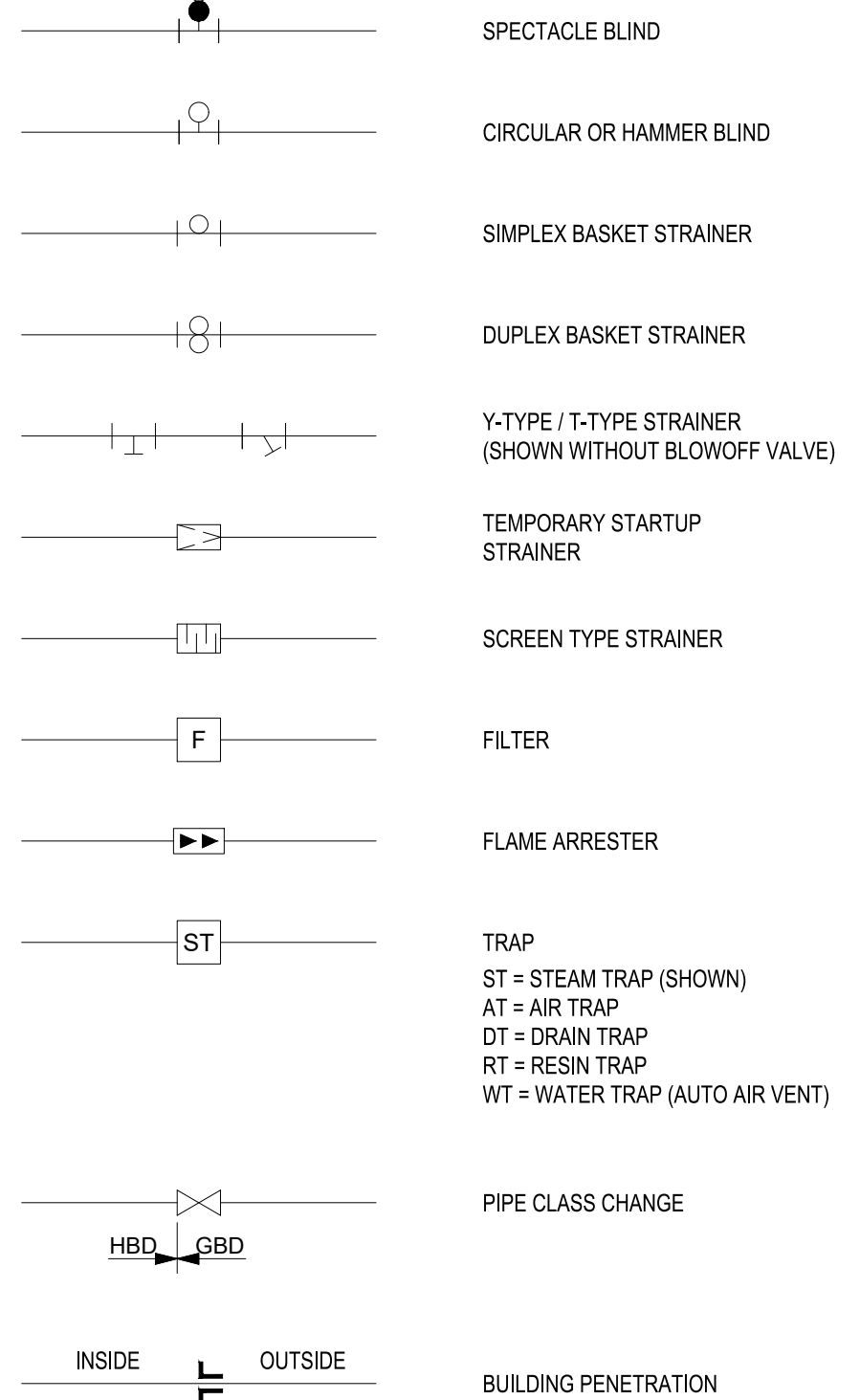
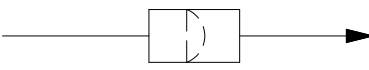
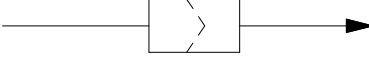
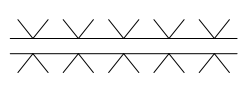
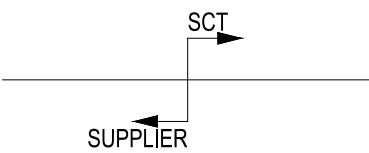
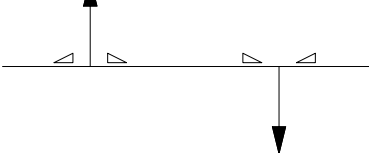


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EQUIPMENT		EQUIPMENT (CONT.)		PROCESS PIPING (CONT.)		LINE VALVES (CONT.)		MISCELLANEOUS (CONT.)		GENERAL NOTES : 1. THIS DRAWING IS PREPARED FOR THE PROPOSAL PURPOSE ONLY AND CAN BE CHANGED ACCORDINGLY IN THE DETAIL DESIGN PHASE. NOTES :																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
 SINGLE FLOW  DOUBLE FLOW  GENERATOR  HORIZONTAL CENTRIFUGAL PUMP  VERTICAL PUMP  RING SECTION CENTRIFUGAL PUMP  VARIABLE SPEED DRIVE  HYDRAULIC COUPLING  POSITIVE DISPLACEMENT PUMP D : DIAPHRAGM P : PLUNGER  U-TUBE  SHELL & TUBE TYPE HEAT EXCHANGER  STRAIGHT TUBE SINGLE-PASS  STRAIGHT TUBE TWO-PASS  PLATE TYPE HEAT EXCHANGER  PRESSURE VESSELS VERITICAL OR HORIZONTAL (TANK, RECEIVERS, DRYERS, SEPARATORS, ETC.)  CONE ROOF TANK  FLOATING ROOF TANK  RECIPROCATING AIR COMPRESSOR  ROTARY AIR COMPRESSOR  MIXER / AGITATOR  CHEMICAL STORAGE DRUM WITH DISPENSING PUMP  PRESSURIZED GAS BOTTLE  FUME SCRUBBER  CO2 ABSORBER  DAMPENER CAUBRATION CYLINDER		 EYE WASHER & SAFETY SHOWER  SAMPLE COOLER  VACUUM DEAERATOR  VACUUM PUMP  EJECTOR OR EDUCTOR  STEAM JET AIR EJECTOR COOLER  DESUPERHEATER		 LOOP SEAL  UPWARD & DOWNWARD SLOPE  VENT  EXHAUST VENT INTO THE OPEN AIR  DIRECT DRAIN ED : EQUIPMENT DRAIN CD : CHEMICAL DRAIN  INDIRECT DRAIN OR FUNNEL DRAIN		 GATE  GLOBE  ANGLE  THREE-WAY  FOUR-WAY  NEEDLE  PLUG 2-WAY 3-WAY 4-WAY  DIAPHRAGM  BALL 2-WAY 3-WAY  BUTTERFLY  HOSE  PINCH  CHECK  FOOT  FLOW LIMITING (OR EXCESS FLOW) CHECK  POWER ACTUATED NON-RETURN VALVE  BACKFLOW PREVENTER  STOP CHECK  ANGLE CHECK  TANDEM BLOW DOWN  MOTOR OPERATED VALVE  SOLENOID VALVE		 CONTROL VALVE WITH POSITIONER  WATER SEAL  BELLOWS SEALED VALVE  PRESSURE BALANCING EXTERNAL PIPE WITH SMALL VALVE  PRESSURE RELIEVING EXTERNAL PIPE WITH SMALL VALVE  PRESSURE BALANCING & RELIEVING EXTERNAL PIPE WITH SMALL VALVE  ACTUATOR G = GEAR (SHOWN) A = AIR WRENCH C = CHAIN X = EXTENSION STEM B = WITH INTEGRAL BYPASS VALVE D = DEEP STUFFING BOX AND LANTERN GLAND T = TAP IN BOTTOM OF BODY (FOR ASME TEST)  		 DEBRIS FILTER  BALL SCREEN  BIRD SCREEN  VACUUM BREAKER  DIAPHRAGM SEAL  VORTEX BREAKER  FLOW GLASS  SEAL PIT  SPRAY NOZZLE OR SPARGER (SHOWN WITH UPWARD/DOWNWARD FLOW)  PROCESS LINE BREAK OR RESUMPTION POINT  SUPPLIER INTERFACE  PIPING HIGH POINT VENT (V) OR PIPING LOW POINT DRAIN (D)  SLIDING GATE  INSTRUMENTATION LINE BREAK OR RESUMPTION POINT  DESIGNATION FOR EQUIPMENT AND PIPING WHICH APPEAR ON OTHER P & I D'S  BOUNDARY OF EQUIPMENT PACKAGE PROVIDED BY SUPPLIER.  DRAIN LEG, DRIP POT		FOR INTERNAL USE THIS DRAWING IS THE PROPERTY OF SAMSUNG C&T CORPORATION IT IS NOT TO BE COPIED OR USED IN ANY WAY DETRIMENTAL TO THE COMPANY. 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8		7		6		5		4		3		2		1					
FIRST LETTER		SUCCEEDING LETTERS													SELF-ACTUATED DEVICES-PRESSURE		GENERAL NOTES : 1. THIS DRAWING IS PREPARED FOR THE PROPOSAL PURPOSE ONLY AND CAN BE CHANGED ACCORDINGLY IN THE DETAIL DESIGN PHASE.		
		SENSING DEVICE		DISPLAY DEVICE			CONTROL DEVICE				MISCELLANEOUS DEVICE								
SYMBOL	MEASURED OR INITIATING VARIABLE	PRIMARY ELEMENT	TRANSMITTER	INDICATOR	RECORDER	INTEGRATING INDICATOR	ARARM ANNUNCIATOR			CONTROL STATION MAN/AUTO	CONTROLLER	CONTROL VALVE	OTHER FINAL CONTROL ELEMENT	SWITCH	LOCAL OBSERVATION GLASS	SOLENOID RELAY OR POSITIONER	TEST POINT	USER'S CHOICE	UNCLASSIFIED
()	TYPICAL SYMBOL	()E	()T	()I	()R	()QI	()AL	()AH	()AHL	()K	()C	()CV	()Z	()S	()G	()Y	()P	()N	()X
A	ANALYSIS	A E	A T	A I	A R		AAL	AAH	AAHL	A K	A C		A Z	A S		A Y	A P	A N	A X
B	BURNER FLAME	B E	B T	B I	B R		BAL				B C			B S		B Y	B P	B N	B X
C	CONDUCTIVITY	C E	C T	C I	C R		CAL	CAH	CAHL	C K			C Z	C S		C Y	C P	C N	C X
D	DENSITY (MASS) OR SPECIFIC GRAVITY	D E	D T	D I	D R		DAL	DAH	DAHL	D K	D C		D Z	D S		D Y	D P	D N	D X
E	VOLTAGE (EMF)	E E	E T	E I	E R		EAL	EAH	EAHL	E K	E C		E Z	E S		E Y		E N	E X
F	FLOW	F E	F T	F I	F R	FQI	FAL	FAH	FAHL	F K	F C	FCV	F Z	F S	F G	F Y	F P	F N	F X
G	GAUGING (DIMENSIONAL)	G E	G T	G I	G R		GAL	GAH	GAHL	G K	G C			G S		G Y		G N	
H	HAND (MANUAL INITIATED)	H									H C		H Z	H S		H Y		H N	
I	CURRENT	I E	I T	I I	I R	IQI	IAL	IAH	IAHL	I K			I Z	I S		I Y		I N	I X
J	POWER	J E	J T	J I	J R	JQI	JAL	JAH	JAHL	J K			J Z	J S		J Y		J N	J X
K	TIME	K E	K T	K I	K R	KQI	KAL	KAH	KAHL		K C		K Z	K S		K Y		K N	K X
L	LEVEL	L E	L T	L I	L R		LAL	LAH	LAHL	L K	L C	LCV	L Z	L S	L G	L Y	L P	L N	L X
M	MOISTURE OR HUMIDITY	M E	M T	M I	M R		MAL	MAH	MAHL	M K	M C		M Z	M S		M Y	M P	M N	M X
O	USER'S CHOICE																		
P	PRESSURE OR VACUUM	P E	P T	P I	P R		PAL	PAH	PAHL	P K	P C	PCV	P Z	P S		P Y	P P	P N	P X
PD	PRESSURE DIFFERENTIAL	PDE	PDT	PD I	PDR		PDAL	PDAH	PDAHL	PDK	PDC		PDZ	PDS					
Q	QUANTITY	Q E	Q T	Q I	Q R	QQI	QAL	QAH	Q AHL	Q K	Q C		Q Z	Q S		Q Y		Q N	Q X
S	SPEED OR FREQUENCY	S E	S T	S I	S R	SQI	SAL	SAH	SAHL	S K	S C		S Z	S S		S Y		S N	S X
T	TEMPERATURE	T E	T T	T I	T R		TAL	TAH	TAHL	T K	T C	TCV	T Z	T S		T Y	T P	T N	T X
U	MULTIVARIABLE			U I	U R								U Z	U S		U Y		U N	U X
V	VIBRATION		V T	V I	V R		VAL	VAH	VAHL	V K	V C		V Z	V S		V Y	V P	V N	V X
W	WEIGHT OR FORCE	W E	W T	W I	W R	WQI	WAL	WAH	WAHL	W K	W C		W Z	W S		W Y		W N	W X
X	UNCLASSIFIED																		
Y	EVENT OR STATE	Y E	Y T	Y I	Y R		YAL	YAH					Y Z	Y S		Y Y		Y N	Y X
Z	POSITION	Z E	Z T	Z I	Z R		ZAL	ZAH	Z AHL				ZZ	Z S		Z Y		Z N	Z X
GENERAL INSTRUMENTS				SENSOR-FLOW				SENSOR-FLOW (CONT.)				SENSORS-PRESSURE				VALVES ACTUATORS			
FF INSTRUMENT FOR SINGLE MEASURED VARIABLE AND ANY NUMBER OF FUNCTIONS. INSTRUMENT IS LOCALLY MOUNTED. INSTRUMENT FOR SINGLE MEASURED VARIABLE AND ANY NUMBER OF FUNCTIONS. INSTRUMENT IS LOCALLY MOUNTED. FF=FOUNDATION FIELDBUS INSTRUMENT IS MOUNTED ON A LOCAL PANEL (TYPICAL) INSTRUMENT IS MOUNTED ON A MAIN CONTROL PANEL (TYPICAL) INSTRUMENT FOR PERFORMANCE TEST ONLY INSTRUMENT FOR PERFORMANCE TEST PLUS NORMAL USE INSTRUMENT IS MOUNTED BEHIND A PANEL (TYPICAL)				FE FE															

(3) SERIAL NO.

NUMERIC SEQUENCE IN EACH SYSTEM
(E.G. 00,01,02, etc.)

EQUIPMENT NUMERIC SEQUENCE

0	BREAKDOWN LEVEL 1					BREAKDOWN LEVEL 2					
U	U	S	S	S	W	W	A	A	N	N	N

	PLANT UNIT	"UU" CHARACTER DESIGNATION
POWER PLANT	POWER PLANT COMMON	0
	FIRST GROUP COMMON (INCL. FIRST GROUP STG)	10
	FIRST GROUP #1 GTG / HRSG	11
	FIRST GROUP #2 GTG / HRSG	12
	SECOND GROUP COMMON (INCL. SECOND GROUP STG)	20
	SECOND GROUP #1 GTG / HRSG	21
	SECOND GROUP #2 GTG / HRSG	22

00LAB11
BROOT
250H LAB1

00LAB11
AA001

00LAB11
AT001

00LAB01
AP001

00LAB21
BROOT
250H LAB2

NOTES :

001 000

PROJECT DWG NO. D00P-00GEN00-M05-0005		SHEET	REV.
		1/1	A
PROJECT NO.	REF. DRAWING NO.	SCALE	REF. CODE
D00P		NONE	

- GENERAL NOTES :
- REFER TO SYMBOL & LEGEND
DWG. NO. D00P-00GEN004-M05-0002--0005
 - THIS DRAWING IS PREPARED FOR THE PROPOSAL PURPOSE ONLY AND CAN BE CHANGED ACCORDINGLY IN THE DETAIL DESIGN PHASE.

NOTES :

- THIS P&ID IS FOR GROUP#10. CONFIGURATION IS IDENTICAL FOR GROUP#20.
- DRAIN POT AND WARMUP LINE SHALL BE LOCATED AS CLOSE AS POSSIBLE FROM STEAM TURBINE MAIN STOP & CONTROL VALVE.
- DRAIN VALVES SHALL BE LOCATED AS CLOSE AS POSSIBLE FROM DRAIN POT.

- * DENOTES EQUIPMENT WILL BE SUPPLIED BY HRS&G.
- ** DENOTES EQUIPMENT WILL BE SUPPLIED BY ST SUPPLIER.

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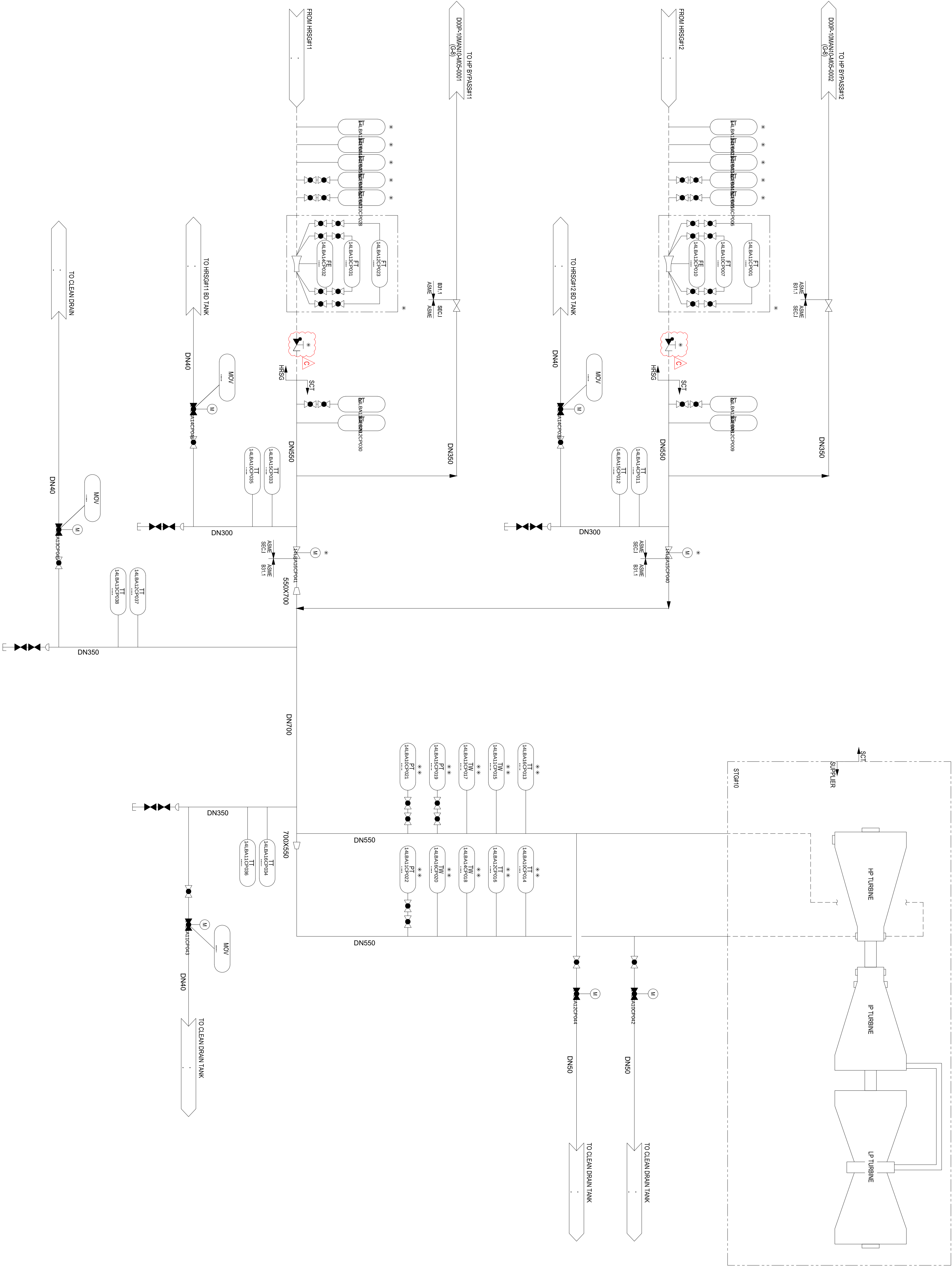
C	24.NOV.2025	FOR INTERNAL USE			
B	01.OCT.2025	FOR INTERNAL USE			
A	19.SEP.2025	FOR INTERNAL USE			
REV.	DATE	DESCRIPTION	PREPARED	REVIEWED	APPROVED
EMPLOYER					

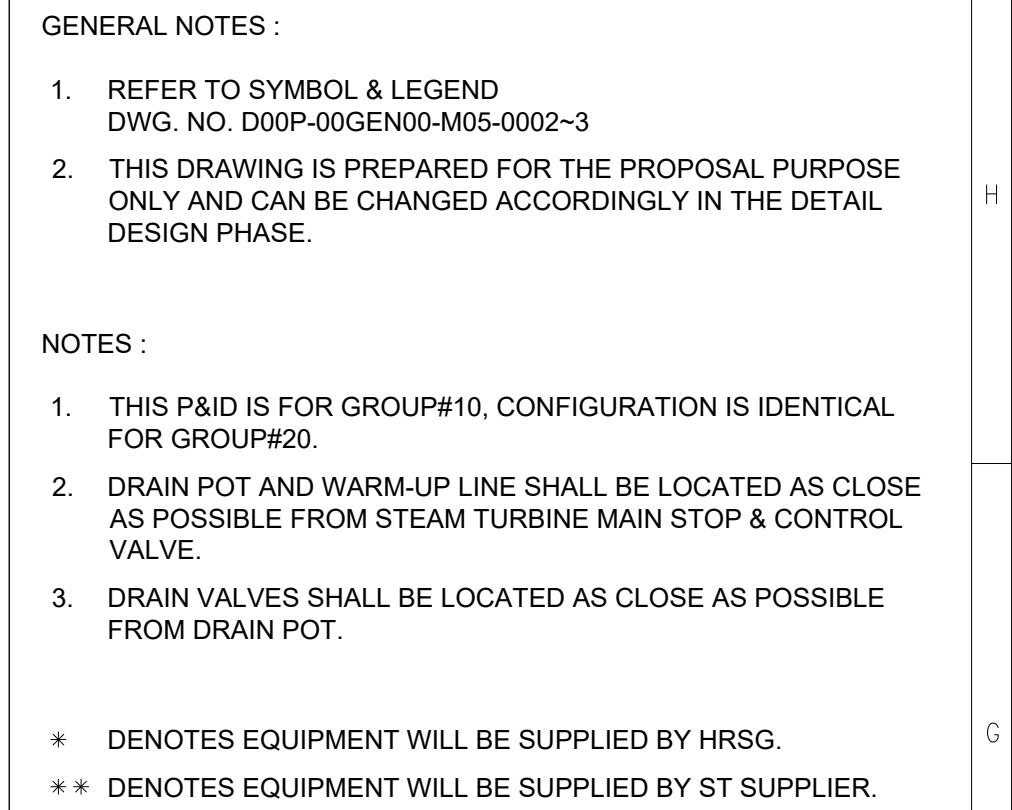
TENDERER
SAMSUNG C&T
Engineering & Construction Group

PROJECT NAME
AL NOUF1 PROJECT

DRAWING TITLE
P&ID FOR HP STEAM SYSTEM GROUP #10

PROJECT DWG NO.	D00P-10LBA10-M05-0001	SHEET	REV.
PROJECT NO.	REF. DRAWING NO.	SCALE	REF. CODE
D00P		NONE	





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TENDERER		 SAMSUNG C&T Engineering & Construction Group	
PROJECT NAME		AL NOUF1 PROJECT	
DRAWING TITLE		P&ID FOR CRH STEAM SYSTEM GROUP 10	
PROJECT DWG NO.		SHEET	REV.
D00P-10LBC40-M05-0001		1/1	A
PROJECT NO.	REF. DRAWING NO.	SCALE	REF.CODE
D00P		NONE	